

# Jonah A. Wilkes

jonah.wilkes@verizon.net | 571-326-9383 | Falls Church, VA · Open to relocation | [LinkedIn](#) | [Portfolio Website](#)

## Summary

---

Aerospace Engineering student at Illinois Institute of Technology pursuing a combined B.S./M.S. in Mechanical and Aerospace Engineering, expected May 2026, with a completed minor in Astrophysics and an engineering curriculum focus in dynamics, controls, and estimation. Experience includes aerospace systems analysis, multidisciplinary design, mission planning, technical evaluation, and research. Skilled in Python and MATLAB for modeling, simulation, signal processing, optimization, and engineering analysis, with project experience spanning spacecraft planning and design, GNSS signal processing, and NASA mission concept development. Background includes systems integration, ConOps development, and trade studies through NASA's L'SPACE programs, as well as technical analysis supporting aerospace and mission-focused engineering projects.

## Experience

---

### **NASA L'SPACE Professional Development Training (MCA) — NASA**

May 2025 – August 2025 | 15 hr/wk

- Served as the team's primary Mechanical Engineer for a Mars rover mission concept, developing a Siemens NX CAD model that integrated mechanical, payload, thermal, power, and communications subsystems.
- Developed the mission Concept of Operations (ConOps) across four mission phases, coordinating with multidisciplinary team members to support system feasibility and mission execution planning.
- Helped define mission and subsystem design under NASA-style project standards, advancing the concept through Preliminary Design Review (PDR).
- Presented the mission concept and design to a Standing Review Board, discussing subsystem integration decisions and the overall mission structure.

### **Aerospace Research Assistant — Armour College of Engineering, IIT**

August 2024 – December 2024 | 15 hr/wk

- Developed design and optimization algorithms in Python to accelerate conceptual aircraft design, specifically for improving on the design of a Boeing 737.
- Built tools to generate and evaluate large parametric design spaces, enabling rapid assessment of performance trends across multiple design variables and configurations.
- Consolidated the analysis and optimization workflow into an interactive application that streamlined trade studies, improved comparison of candidate designs, and reduced design cycle time.
- Awarded First Place at the Fall 2024 Armour R&D Expo for this research project.

### **Aerospace Research Assistant — Armour College of Engineering, IIT**

May 2025 – Current | 10 hr/wk

- Developed MATLAB based processing methods for Global Navigation Satellite System Reflectometry (GNSS-R) data collected over Antarctic glacial surfaces through a NASA Illinois Space Grant-funded research project.
- Processed and evaluated raw reflected satellite signals to extract signal-to-noise metrics and characterize surface reflection behavior using signal processing techniques.
- Performed technical analysis of reflected-signal datasets to support interpretation of glacial surface conditions and ongoing aerospace remote sensing research.
- Contributing results to an upcoming peer-reviewed publication.

### **NASA L'SPACE Professional Development Training (NPWEE) — NASA**

May 2024 – August 2024 | 10 hr/wk

- Contributed to a NASA style project proposal within a multidisciplinary team, working under NASA proposal standards and contributing to the project's technical development.
- Developed substantial mission design, technical analysis, and proposal content for a complete NASA-style mission proposal reviewed through official NASA channels, helping define mission concept content and written technical material.
- Participated on a NASA-style proposal review board, evaluating peer submissions and providing technical feedback on mission design approaches.

## Astrophysics Research Assistant — Department of Physics, IIT

May 2024 – Current | 15 hr/wk

- Analyzed stellar spectra in a Python environment to derive radial velocity curves for 98 binary systems, extract orbital parameters, and evaluate system dynamics.
- Developed Monte Carlo simulations to generate synthetic stellar populations and compare modeled distributions against observed data, performing statistical evaluation of binary-star formation hypotheses.
- Primary author on a forthcoming publication from this research, investigating binarity in stellar systems through an analysis of Sub-Subgiants.

## Education

---

*Fall 2021 – Present*

**Illinois Institute of Technology**

*Bachelor of Science in Aerospace Engineering*

*Master of Science in Mechanical and Aerospace Engineering*

*Minor in Astrophysics*

*Chicago, IL*

*ABET Accredited*

*ABET Accredited*

*Completed Fall 2024*

**Cumulative GPA:** 3.840/4.0

**Expected Graduation:** May 2026

**Armour College of Engineering Dean's List:** Every semester enrolled (Fall 2021 – Spring 2025)

**IIT STEM Scholarship Recipient**

**Relevant Courses:** Spacecraft Dynamics & Design, Aircraft Design, Aerospace Propulsion, Navigation Systems, Optimal State Estimation, Optimal Control, Systems Analysis and Control, Advanced Dynamics, Aerodynamics, Aerostructures, Compressible Flow, Computational Mechanics.

## Technical Skills

---

**Programming & Analysis:**

Python, MATLAB, Linux, Monte Carlo simulation, statistical analysis, data visualization, signal processing, optimization, technical modeling

**Aerospace Modeling & Simulation:**

STK, Simulink, AVL, XFLR5, OpenRocket, spacecraft dynamics modeling, orbital and mission analysis, Lambert solvers, porkchop plot analysis

**Systems Engineering & Mission Design:**

Systems integration, ConOps, trade studies, mission planning, design reviews, requirements development, sensitivity analysis, multidisciplinary design

**CAD & Design Tools:**

Siemens NX (Certified), SolidWorks, 3D assemblies, mechanical drawings

**Controls, Estimation & Navigation:**

Optimal state estimation, optimal control, navigation systems analysis, dynamics and control modeling

## Extracurriculars

---

**Illinois Tech Baseball**

Fall 2021 – Present

5 year varsity athlete and catcher; developed leadership, teamwork, and communication in a competitive team environment

**Illinois Tech Rocketry Club**

Fall 2022 – Present